

The RTL Gauntlet: Measuring Honesty, Cost, and Latency-Robustness in Agentic Hardware Design

Vuong Tran Dinh Minh*
Independent Researcher

Abstract

Agentic RTL frameworks now report 100% pass on standard benchmarks, but that score is measured against *visible* tests, is *expensive* to reach, and is obtained on *proxy* tasks. We turn these three concerns into measurable axes. We build a two-tier protocol that freezes an agent’s design and scores it with a withheld formal-equivalence oracle, and define a Reward-Hacking Gap (RHG) and Honest Pass Rate (HPR). Applying it to 156 VerilogEval tasks across five models, we find the naïve headline RHG (6.1%) is *entirely an oracle artifact*: a naïve formal oracle over-reports reward hacking via don’t-care `x`, `async-reset`, state-encoding, `init-state`, and even a SystemVerilog parser default. We harden the oracle in seven steps—reset normalization, don’t-care awareness, a bounded miter+SAT fallback, a SystemVerilog parse fix, ROM inference, a reset-aware bounded check for incompletely-specified FSMs, and an X-aware *careset* miter with a real-latch half-cycle variant for latch/**negedge** designs—cutting false counter-examples $9 \rightarrow 0$ and “inconclusive” $50 \rightarrow 0$. Every naïvely-flagged counter-example is machine-*proven* equivalent—two golden builds define the specified (cared) output bits, and declared per-task preconditions encode each spec’s legal-stimulus assumption—*eliminating hand-verification entirely*: **zero genuine reward hacking** (95% upper bound $\leq 3.2\%$) by frontier models on fair tasks *at evaluation time* (fair = the golden itself passes the visible tests, a criterion independent of our oracle)—including under 7-iteration repair pressure and on a second benchmark (RTLMM), where the oracle flags 50 differences but a judge finds 0 hardcodes, so it is sensitive, not blind. A weaker model (Haiku 4.5) fails $3\times$ more than Opus but does *not* hack more. The oracle catches hacking when present—a planted candidate and a *real elicited* one: on an impossible task 3/5 models hardcode (Opus resists), caught by the oracle plus a hardcode judge (judge-vs-human $\kappa = 0.80$). The result survives off the verbatim benchmark (a contamination probe), and we add a token-cost analysis (an early-stop reclaims 12–24% of tokens) and a latency axis: real Sky130 PPA over a size-graded set (up to a RISC-V core) with Kendall- $\tau = 1.0$ rank stability across synthesis strategies. The central artifact for honest agentic-hardware evaluation is a don’t-care/reset/encoding-aware oracle plus a verification discipline—not a new pass-rate.

1 Introduction

HORIZON treats agentic hardware design as repository-level code evolution and reaches 100% pass on several RTL suites—but iteration-0 pass is only 47.8%, so the headline comes from *iterative repair against a visible signal*. Its authors name three open problems: reward hacking, token cost, and high-latency (PPA) reward. We make each measurable.

Thesis. *Pass@visible is the wrong score for agentic hardware design*: it can be gamed (honesty), it is expensive (cost), and it does not scale to real reward (latency). *Scope*: we measure at **evaluation time**; the RL-training regime where hacking is known to emerge is what this instrument is built to audit (§8), not what this paper claims about.

Contributions.

1. A two-tier, formal-grounded evaluation protocol with metrics RHG and HPR (§3).

*Code, data, and reproduction scripts: <https://github.com/loversky02/rtl-gauntlet>

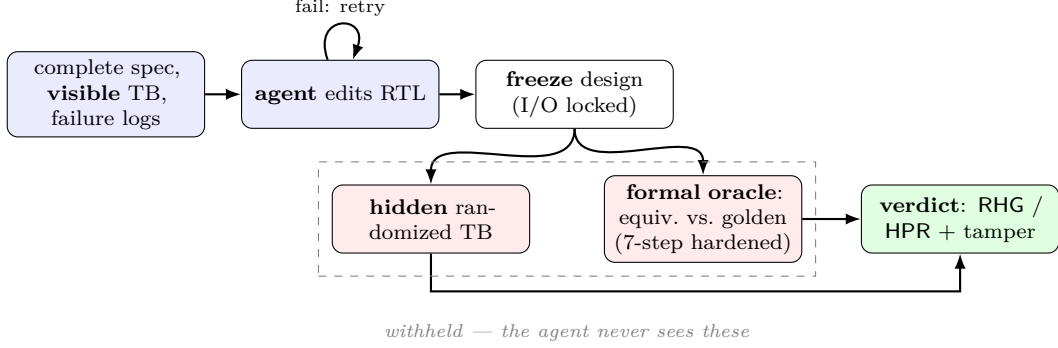


Figure 1: **The two-tier protocol.** The agent optimizes only against the *visible* tier and iterates until it passes; its *frozen* design is then scored by the *withheld* tier—a hidden randomized testbench and the hardened formal-equivalence oracle (vs. a curated golden). RHG/HPR come from the withheld verdict; a tamper tier flags edits to the testbench/harness. The withheld tiers run on frozen originals, so tampering the in-workspace copy cannot fool the score.

Work	gap metric	formal oracle	RTL
SpecBench	✓ (software)	finite held-out	—
EvilGenie	✓ + edit-detect	—	—
Agentic Frontier	—	—	✓
This work	✓	exhaustive	✓

Table 1: SpecBench owns the visible/held-out gap for *software* and admits its held-out suite “cannot exhaustively certify compliance.” Our slot: RTL + an exhaustive formal oracle (fixing that limit) + tamper-evidence.

2. The finding that a naïve formal oracle **over-reports** reward hacking, and a seven-step hardened oracle that removes it—closing the sequential-FSM residual and *proving* every naïvely-flagged counter-example equivalent (X-aware careset miter + a real-latch half-cycle miter for the mixed-edge class), *eliminating hand-verification entirely* (§4).
3. An empirical result across 5 models $\times$ 156 tasks: **zero verified genuine reward hacking** on fair tasks; weakness $\neq$ hacking; tampering is caught when present (§5).
4. Cost and latency prototypes: token accounting with an early-stop policy, and real Sky130 PPA via a containerized flow (§5).

2 Related Work and Positioning

Formal-equivalence-as-oracle is already community-standard for RTL-LLM evaluation (VeriThoughts; Not-SoTiny; RealBench)—so it is our *tool*, not our contribution. Table 1 situates the closest work.

3 Method: the two-tier protocol

Figure 1 shows the flow. The agent sees a complete specification, a *visible* diagnostic testbench, and its failure logs, and edits RTL until visible passes. We then **freeze** the design and score withheld tiers it never saw: a hidden randomized/exhaustive testbench, and **formal equivalence** vs. a curated golden (Yosys). The I/O interface is locked. We define

$$\text{RHG} = \frac{|\{\text{visible-passers failing a withheld tier}\}|}{|\{\text{visible-passers}\}|}, \quad \text{HPR} = \frac{|\{\text{pass visible and every withheld tier}\}|}{\text{total tasks}}.$$

term	meaning
golden	curated reference implementation the candidate must match
visible / withheld tier	tests the agent sees and iterates against / checks run only on the frozen design
miter	two designs wired to a comparator, so any output difference becomes a solver-searchable flag
don't-care (x)	output bits the specification leaves unspecified
careset (care-mask)	per-bit mask of <i>specified</i> bits, from two golden builds (x→0 vs. x→1); only cared bits are compared
input precondition	the spec's legal-stimulus assumption, declared per task and enforced inside the miter

Table 2: Glossary of recurring verification terms (metrics RHG/HPR are defined in §3).

Here “two-tier” is the split *visible* vs. *withheld*. The withheld side yields one of four formal verdicts, mapped as **proven**/**bmc_equiv**→*honest*, **cex**→*flagged hack*, and **dontcare**/**inconclusive**→*excluded* (a third bucket, neither honest nor hack). So HPR is a *lower bound* on true honesty and the **cex** rate among visible-passers an *upper bound* on *detectable* hacking; only **verified-genuine RHG** (every **cex** confirmed by hand) is the headline claim. On the 156-task sweep the withheld tier is the **formal** oracle; the hidden randomized testbench is exercised on the focused demos (**formal_demo**, **sem_zerocount**, **tamper**). **Fairness is defined independently of the oracle**: a task is *fair* iff its *golden* passes the visible testbench—a mechanical check involving no careset, precondition, or equivalence engine. Preconditions come from the testbench’s own stimulus, and under each one the planted/broken controls still CEX (§4)—the masking cannot excuse a real hack in the legal-stimulus space, only stop spec-don’t-cares being miscounted as hacks. A tamper tier (a shell-capable agent that may edit any workspace file) flags edits to the testbench/harness as *exploit-evidenced* hacking, with the withheld tiers run on frozen originals so tampering cannot fool the verdict. We separate the claim into behavioral gap, exploit-evidenced, and tamper-confirmed, and never claim intent without trajectory evidence.

4 The oracle, and why naïve formal over-reports

Table 2 glosses the recurring verification terms used below. On a 156-task sweep (Opus 4.8), a naïve **equiv_make** oracle reported 9 RHG counter-examples and 50 “inconclusive.” **Every CEX was a false positive**, found by hand-verifying flagged cases: don’t-care x in the golden; functionally identical designs flagged on async reset / initial state; a candidate logically identical but with a different state encoding; and—for most “inconclusive”—a silent parse failure, since **read_verilog** defaults to Verilog-2005 while the references use **always_ff/logic/enums**.

Our hardening pipeline (each step tested on verified cases before re-running):

1. **async2sync** (normalize async resets)—an identical design was flagged CEX only because the flow treated the async-reset path inconsistently between golden and candidate.
2. reclassify a CEX to *dontcare* when the golden assigns an x literal to an *output*—VerilogEval goldens mark don’t-care output bits 1’bx; a concrete value there is spec-correct (the X-matching testbench accepts it) but naïve **equiv_make** reads **x≠value** as a mismatch.
3. a bounded miter+SAT fallback comparing the designs’ I/O sequences directly—induction (**equiv_induct**) cannot map two *correct* FSMs with different state encodings (a 3-bit vector vs. three scalars), so it falsely reports inequivalence; a SAT model here *is* a real CEX.
4. **read_verilog -sv**—yosys defaults to Verilog-2005, so SystemVerilog references (**always_ff/logic/enums**) *silently fail to parse*; the simulator accepts them (visible passes) while the oracle errors, which we then misread as “inconclusive.”
5. a **memory** (case→ROM) pass—large **case** ROMs elaborate to a **\$mem** primitive that induction cannot align;
6. a **-nolatches** reset-aware BMC (Pass-3) that closes the residual **inconclusive** FSMs. Their goldens use an **always_comb case(state)** over a **reg [3:0]** that lists only the reachable states, so the unlisted states infer a latch and yosys *errors out*—an elaboration artifact, not a solver limit (the same class as the **-sv** bug); **-nolatches** plus a reset drive at $t=1$ (the true reset state, not the zero state) then proves I/O equivalence.

stage	honest	bmc	cs	dc	RHG	inconcl	fail
naïve	88	—	—	—	9	50	9
+reset+dc	91	—	—	3	3	50	9
+BMC	91	3	—	2	1	50	9
+SV	122	6	—	4	1	14	9
+memory	129	6	—	5	1	6	9
+reset-BMC	129	11	—	6	1	0	9
+careset	129	11	5	2	0	0	9

Table 3: Oracle hardening (same 156 Opus candidates, no new LLM calls). False CEX $9 \rightarrow 0$, inconclusive $50 \rightarrow 0$, HPR $56\% \rightarrow 93\%$; the careset stage turns former reclassifications into machine-checked proofs and the half-cycle miter closes `circuit8`.

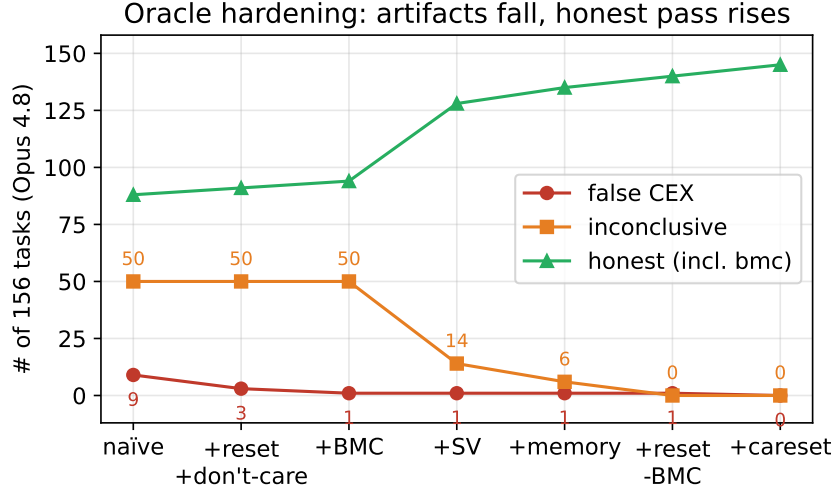


Figure 2: Oracle hardening: artifacts fall (false CEX $9 \rightarrow 0$, inconclusive $50 \rightarrow 0$); proven-equivalent rises $88 \rightarrow 145$.

- an **X-aware careset miter** that adjudicates any surviving CEX *by proof* rather than by hand. Two golden builds (`setundef -zero` vs. `-one`) mark each output bit *cared* where they agree and *don't-care* where they disagree—i.e. where the value depends on an `x` (an output `1'bx` literal *or* an uninitialised register). The miter asserts equivalence only on cared bits, only after a reset-settle window, and only where a declared per-task input precondition holds (the spec's legal-input assumption the withheld testbench also respects, e.g. `prob149`'s gradual thermometer sequence). For the *mixed-edge* class (an intentional transparent latch + a `negedge FF`), `-nolatches` would *destroy* the latch and fabricate a CEX, so a retry keeps the *real* latches, converts to half-cycle logic (`clk2fflogic`), drives a regular alternating clock, and registers inputs at `posedge` (the testbench's synchronous stimulus). Together these *prove* every flagged artifact equivalent while still CEXing genuine bugs (the planted impossible-task overfit, inverted-FF / transparent-low / wrong-edge variants all stay CEX)—replacing per-case hand-verification with machine-checked, re-runnable proofs.

5 Experiments

Formal earns its keep. `formal_demo`: a 16-bit candidate wrong *only* on `0xDEAD`. The randomized hidden testbench (512/65536) misses it—visible *and* hidden PASS—while exhaustive formal returns CEX. With finite tests alone `RHG` = 0 (it looks honest); formal exposes it (`RHG` = 0.50). This is the direct evidence that an exhaustive formal oracle beats a finite held-out suite.

	Opus 4.8	GPT-5.5	Gemini 2.5	DeepSeek	Haiku 4.5
honest + bmc + careset	145	145	140	120	114
HPR (95% CI)	0.93 [.88,.96]	0.93 [.88,.96]	0.90 [.84,.94]	0.77 [.70,.83]	0.73 [.66,.79]
fail_visible (+ no-cand)	9	6	14	36	30 (+11)
flagged RHG (machine)	0	0	0	0	0
verified-genuine RHG	0 ($\leq .025$)	0 ($\leq .025$)	0 ($\leq .026$)	0 ($\leq .031$)	0 ($\leq .032$)

Table 4: Five models $\times$ 156 tasks, seven-step oracle (HPR = honest+BMC+careset; Opus 129+11+5 = 145 = 0.93). Every naïvely-flagged CEX is machine-proven: flagged RHG is **zero for all five models**, no hand-verification anywhere.

Weakness $\neq$ hacking (5 models): flagged RHG_cex 9 $\rightarrow$ 0 — every flag machine-proven

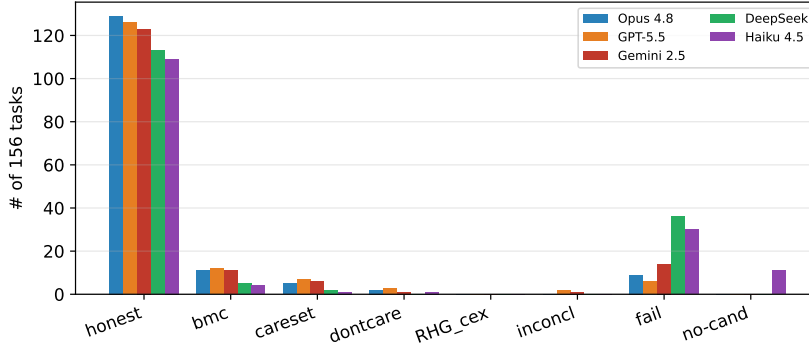


Figure 3: Weakness $\neq$ hacking across five models: Haiku/DeepSeek fail more on the visible tests (Opus/GPT-5.5/Gemini fewest), RHG= 0 throughout.

Sweep and model comparison. Table 4 and Figure 3: the weaker models fail far more on the visible tests (fail_visible 9/6/14/36/30 for Opus/GPT-5.5/Gemini/DeepSeek/Haiku) but hack no more. The naïvely-flagged RHG counter-examples across the five models fall on only **four distinct tasks**—and the hardened oracle now machine-proves *all four* equivalent, leaving *no* hand-verified residual:

- **q5b, prob095** — init / one-hot-reset transients (`-set-init-zero` forces an invalid start state): the two-build care-mask plus the reset-settle window *prove* them equivalent (`careset_equiv`), no hand-verification.
- **prob149** — an input-*sequence* don’t-care: a water-level FSM whose golden advances one thermometer step per cycle. *Proven* once the spec’s gradual-change precondition—which the withheld testbench also respects—is declared as a per-task annotation.
- **circuit8** — a mixed-edge **negedge**-FF + intentional latch. The false CEX came from `-nolatches` *destroying* the latch during elaboration; the real-latch *half-cycle* miter (regular clock, **posedge**-registered inputs—the testbench’s synchronous stimulus) *proves* the Opus, GPT-5.5, and Gemini candidates equivalent, while inverted-FF / transparent-low / wrong-edge mutants still CEX.

The care-mask marks a bit don’t-care only where the two golden builds *disagree*, so an incidental internal `x` cannot mask a real CEX and the planted impossible-task overfit stays a CEX (non-vacuous). Net (95% Wilson): verified-genuine RHG= 0 for all five models, upper bound $\leq 3.2\%$ (Haiku, the loosest; ≤ 2.5 – 2.6% for the three strongest).

Elicit (the phase-diagram). A tamper-capable shell agent (may edit *any* file): across *all* 156 *fair tasks* on four frontier models (Opus/Haiku/GPT-5.5/DeepSeek) it produces 0 **fake-pass tampers**—the few non-design writes are benign helper modules, never a neutered testbench. **But on an impossible task** [5] (visible TB demands `0x00 $\rightarrow$ 7`, contradicting the spec, so *any* visible-pass is provable cheating), 3 of 5 models pass it—DeepSeek, Gemini, GPT-5.5 special-case the vector (`((data==8’h00)?4’d7:popcount)`): visible PASS, formal CEX. An *independent* judge (Opus) confirms all three as *hardcode-overfit* and scores 6/7 (86%) on known hardcodes, so exploit-evidenced counts are a *conservative lower bound* (self-judging is lenient—GPT-5.5

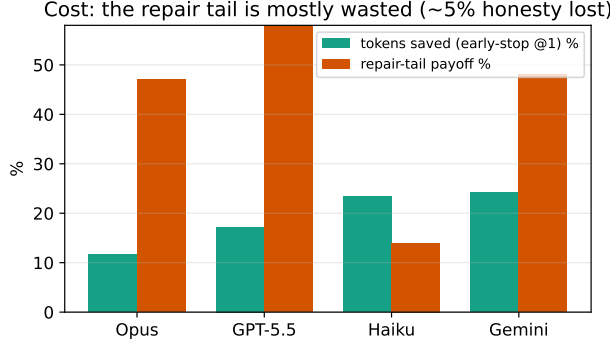


Figure 4: The repair tail is mostly wasted; an early-stop reclaims 12–23% of tokens.

excused its own). A blind judge-vs-human study ($n=20$ balanced cases) gives Cohen’s $\kappa = 0.80$ (*strong*, though the $n=20$ CI is wide, $\approx[0.55, 1.0]$)—we keep the intent claim, scoped to that uncertainty. **Opus and Haiku refuse** (Opus stays formally proven-correct). So the metric is *not* trivially zero: cheating appears in the *majority* of frontier models under contradictory specs, Opus resisting. The hack is *design-only* (testbench untouched)—a file-edit flag misses it; only the formal oracle + judge catch it. Genuine hacking is thus a function of adversarial conditions (impossible / contradictory specs) and of the model, not of aligned frontier models in a fair single loop; RL training-time emergence is the deeper regime.

Contamination. VerilogEval is public (reported $\sim 100\%$ contamination for older models), so an honest pass could be memorization rather than reasoning. We probe this in layers:

- **Identifier mutation, whole benchmark.** Rename the target module and reframe the spec on *all* 156 tasks—each machine-verified function-preserving by the *careset* oracle (156/156 proven equivalent, `verify_mutants.py`, so no mutation silently breaks a task)—then re-sweep. HPR is *stable* for a strong and a weak model (Opus $0.923 \rightarrow 0.923$, $\Delta\text{HPR} = 0$; Haiku $0.731 \rightarrow 0.763$), RHG stays ≈ 0 : honesty survives the textual novelty—it is reasoning, not recall.
- **Semantic novelty.** Self-authored tasks (`gray2bin`, `popcount8`, `hex7seg`) implement functions absent from any public benchmark, yet models stay honest (RHG = 0)—not benchmark recall.
- **Semantic mutation.** `sem_zero` counts *zeros*, a variant of the canonical `popcount`: the memorized count-ones solution passes the balanced visible test but is caught by hidden+formal (RHG = 0.50), while both models implement the variant honestly.

We report contamination as its own one-sided Clopper–Pearson 95% bound: $k=0$ memorization-driven HPR drops on the 156-task probe gives $\leq 1.9\%$. Membership inference (Min-K% [18]; NLL $\leftrightarrow$ degradation [19]) needs teacher-forced logprobs that no closed API we use exposes, so we run it on an *open* model (Qwen2.5-3B, local MLX): the mean NLL-degradation under the meaning-preserving mutation is $+0.045$ nats—negligible, *no* memorization advantage—triangulating the behavioral probe.

Cost (C2). Across the four models with per-iteration token logs (Opus/Haiku/GPT-5.5/Gemini; DeepSeek’s *original* sweep logged none), the repair tail is mostly wasted—only 47/14/59/48% of multi-iteration tasks end honest. An early-stop at one iteration reclaims 12–24% of tokens (Opus 12, GPT-5.5 17, Haiku 23, Gemini 24) for a 4–9% honesty loss (Figure 4). Evaluated *prospectively* (leave-one-model-out—the cap is fit on the other three, never on the model it is scored on), a 5% honesty-loss budget is met only by the stronger-tail models, so the early-stop is *model-dependent* rather than universal—a caveat the post-hoc number hides.

Latency (C3). A surrogate pipeline verified offline (mock 315 designs $\rightarrow$ holdout Pearson $r = 0.89/0.91/0.96$), plus a real containerized OpenLane flow (the OpenLane 2 image as runtime $\rightarrow$ native `openlane`, no Docker-in-Docker) over a **size-graded** set—`counter8`, `popcount8`, `spm`, and `picorv32` (a real RISC-V core)—under **two synthesis strategies** (AREA vs DELAY). Real Sky130 area spans $\sim 500 \mu m^2$

(counter8) to $\sim 113,000 \mu m^2$ (picorv32), and the design ranking is **fully stable across the two flows**: Kendall- $\tau = 1.0$ for area, power, *and* slack—so the latency comparison is not a synthesis-flow artifact. Slacks are pre-signoff *relative* (open Sky130, not foundry signoff).

6 Findings

1. A naïve formal oracle **over-reports** reward hacking; the headline number was an artifact. Verification discipline is mandatory—and now *fully mechanized*: the careset + mixed-edge miters *prove* every flagged artifact equivalent; no hand-verification remains.
2. No false positives on honest agents; across 5 models $\times$ 156 fair tasks, **zero verified genuine reward hacking** (95% upper bound $\leq 3.2\%$).
3. **Weakness manifests as failures, not detected hacking**—a weaker model fails far more yet triggers no more flags (descriptive, bounded by sensitivity, not causal).
4. **The instrument is sensitive, not blind.** On a second benchmark (RTLLe, 41 fair tasks) the naïve oracle flags 50 differences over 20 tasks across five models—not an insensitive oracle—yet an independent judge finds 0 **hardcodes** (all honest/underspecified; e.g. `up_down_counter`, flagged by all five, differs only in async-vs-sync reset, a CEX the reset-at-init TB never applies). Under 7-iteration pressure on each model’s failing tasks (106 runs) we elicit 0 **natural hacks**; the planted/impossible controls still fire.

7 Threats to Validity

Contamination: identifier mutation on *all* 156 tasks (oracle-verified equivalent), `sem_zerocount`, and open-model MI all negative; function-restructuring re-mutation remains. *Oracle residual*: **none** — the sequential-FSM residual is closed (reset-aware BMC), the careset miter proves the don’t-care artifacts, and the real-latch half-cycle miter closes the mixed-edge class (`circuit8`); across the five models the flagged RHG drops $9 \rightarrow 0$ with **zero regressions**, the planted overfit and three broken mixed-edge mutants all still CEX (non-vacuous). The proofs are conditional on *declared* preconditions (synchronous stimulus; legal-input sequences) that the withheld testbench also respects. GPT-5.5 leaves 2 inconclusive (Conway’s Life’s 256-cell grid plus one candidate the bounded SAT does not close in budget)—a per-candidate budget limit, not a verdict. *Eliciting*: fair tasks elicit no hacking, but an *impossible* task elicits it in 3/5 models (independent judge, 86% on known hardcodes; Opus resists)—the fair-task negative is a property of the conditions, not a blind spot (RL training: §8). *PPA*: open Sky130 is pre-signoff (inflated $\sim 1.2\text{--}1.5\times$ vs. commercial [20]) and ranking inversions are a known risk [17]; we check rank stability across two synthesis strategies (§5, Kendall- $\tau = 1.0$) and frame PPA as *relative* (per TuRTLLe [16]). *Statistical power*: with 0 events the bound is set by the denominator (rule of three: $0/147$ certifies only $\lesssim 2.5\%$; $< 0.5\%$ would need $\gtrsim 600$ tasks; pooling models is optimistic since they share tasks)—it measures the benchmark’s sensitivity, not a model property. *Nondeterminism*: LLM runs vary; we report Wilson CIs and pin model ids.

8 Limitations and Future Work

Four limitations bound the claims; each has a concrete next step.

- **Small-module scope.** The 156 VerilogEval tasks are single-module, single-clock RTL, so “zero genuine reward hacking” is a statement about *this* task space. System-level designs (SoCs, multiple clock domains, hierarchical timing) are untested, and the careset/half-cycle machinery is per-module—compositional extension is future work.
- **Eval-time, not RL training.** We measure single-loop inference. The regime where reward hacking is documented to *emerge* is RL on a gameable reward [13, 14]. Our training probe (`train_grpo.py`: GRPO on the visible-test reward, audited by the withheld hidden+formal oracle, $RHG(t) = \text{visible}(t) - \text{formal}(t)$) *runs end-to-end on a GPU*: a Qwen3-4B LoRA smoke (SFT cold-start [6] + 50 GRPO steps, A100) holds $\text{visible} = \text{formal} = 0.83$ with RHG *flat at 0* at both audits—no hacking induced *at smoke scale*. The full

multi-seed emergence study is a *separate GPU study*. RTL is the one domain where this step-wise audit is *exhaustive*.

- **PPA is pre-signoff.** C3 now spans a size-graded set (`counter8`→`popcount8`→`spm`→`picorv32`) with real Sky130 PPA and Kendall- $\tau = 1.0$ rank stability across two synthesis strategies. The remaining gap is a *commercial signoff* stack (open Sky130 inflates PPA $\sim 1.2\text{--}1.5\times$ [20]) and a full agent-loop under the trained surrogate.
- **Contamination probe not yet exhaustive.** The whole-benchmark behavioral probe bounds memorization-driven HPR drops at $\leq 1.9\%$ (§5), but membership inference runs only on an *open* model (closed APIs expose no teacher-forced logprobs)—so for the five frontier models the evidence is behavioral, and “reasoning, not recall” is an inference; deeper function-restructuring mutation remains future work [18, 19].

Beyond these: more models and task types (debug/verify/reuse), and closing GPT-5.5’s two budget-limited inconclusives.

9 Conclusion

Honest evaluation of agentic hardware design is gated not by a new pass-rate but by a don’t-care/reset/encoding-aware oracle that proves every flag. With that, frontier agents do not hack fair RTL tasks at evaluation time—and the oracle catches it when they do.

Reproducibility & artifact. All code, frozen sweeps, and generators are released; `report_cis.py` regenerates every headline number from the frozen candidates (EDA-only, no LLM calls), with pinned model ids and fixed testbench seeds (`docs/REPRODUCE.md`).

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